

Predictive Model Plan – Delinquency Risk Prediction

1. Model Logic (Generated with GenAI)

Selected Model: Logistic Regression

Model Options Considered:

- Logistic Regression – Transparent and regulator-friendly.
- Decision Tree – Easy to interpret but may overfit.
- Random Forest – Higher accuracy but lower interpretability.

Recommended Model: Logistic Regression.

Model Workflow:

Load data → Handle missing values → Encode categories → Select features → Train model → Generate probability scores → Classify risk → Evaluate performance.

Top 5 Input Features:

1. Missed Payments
2. Payment History (Month_1–Month_6)
3. Credit Utilization
4. Debt-to-Income Ratio
5. Credit Score

Model Summary:

Logistic regression predicts the probability that a customer will become delinquent. It is suitable because delinquency prediction is a binary classification problem and the model provides transparent, explainable outputs.

2. Justification for Model Choice

Logistic regression was selected because it balances predictive performance, transparency, and ease of deployment. Financial institutions require explainable models that can justify risk decisions. While more complex models may provide slightly higher accuracy, logistic regression clearly shows how each variable contributes to risk, making it appropriate for Geldium's business and regulatory requirements.

3. Evaluation Strategy

Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- AUC-ROC

Interpretation:

Recall helps identify high-risk customers, precision reduces false alarms, F1 Score balances both measures, and AUC-ROC evaluates the model's ability to distinguish delinquent from non-delinquent customers.

Fairness Checks:

- Compare results across employment groups.

- Monitor false positive and false negative rates.
- Exclude protected characteristics.
- Conduct regular bias reviews.

Ethical Considerations:

The model should support human decision-making, remain transparent, and be regularly monitored to ensure fair treatment of customers.